LIBERO-SMART

120 tasks **6,000 evaluation instances**

Instruction following over familiar manipulation skills

(a) Semantic

Fine-grained semantic following

40 tasks **2,000 instances**

Pick up only the bowl.
Keep holding it.



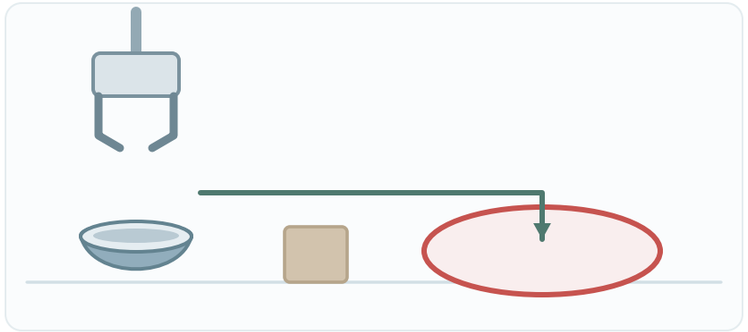
Rewrites, subtasks, atomic motions
Object, spatial and order constraints
Task-specific stopping and holding

(b) Visual Prompt (Vis)

Multimodal instruction understanding

40 tasks **2,000 instances**

Place the bowl inside
the red circle.



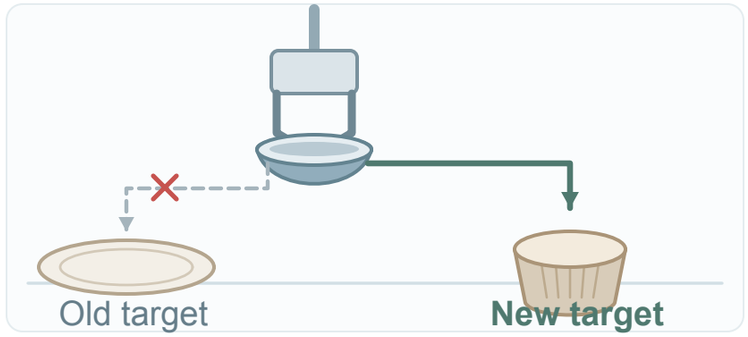
Circle Region Point Path

(c) Control

Mid-execution corrective control

40 tasks **2,000 instances**

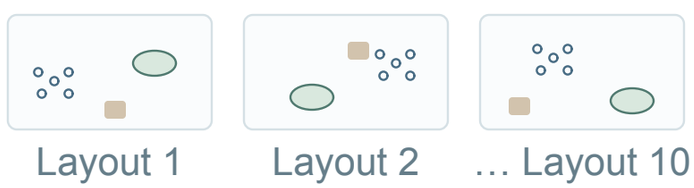
No, place it in the
ramekin instead.



Target correction
Process re-planning
Matched intermediate states

(d) Initialization coverage

10 macro layouts × 5 local variants



50 initializations / task

Control: matched intermediate states

(e) Trajectory-level contracts

Correct target G ∧ **Valid process H** ∧ **Stable result R**

Target and relation | Restrictions and order | Release, hold and stability

Direct policy Strict success for 30 subsequent action steps

Agentic system Strict contract at termination

Goal-only is reported separately as an auxiliary score.

Held-out task combinations, instructions, object-target relations and layouts

Evaluation trajectories are excluded from adaptation; comparisons use matched instances and budgets within each protocol.